

Predicting the inclination of the Gubbio wall from the extended sensor package

Station 02, 21 February 2025 to 10 August 2026, hourly grid

Study report · studies/inclination_prediction/

12 August 2026

Abstract

The instrument package installed at station 02 on 21 February 2025 records air temperature, relative humidity, wall temperature, solar radiation, battery voltage and inclination on one logger. This report asks what that package buys as a *prediction* system. Three questions are answered: which single channel best predicts the inclination, whether a combination does better and how the best combination should be diagnosed, and how far ahead NeuralProphet can forecast with a defensible statement of uncertainty. Every driver is given the chance to act through a transport delay and a thermal inertia before it is ranked, with the delay bounded at twelve hours on physical grounds, and the ranking is out-of-sample throughout.

The headline results are that air temperature dominates every ranking; that on the long summer block the pair `tair + sr` reduces the error by 27 % over air temperature alone, while wall temperature appears in no winning subset in either tier once implausible delays are excluded; that the useful forecast band is roughly three to twelve hours ahead, outside which the model does no better than repeating the previous day; and that the nominal 90 % prediction interval of the deployable configuration actually covers 54–67 % of observations, so it must not be quoted as ninety.

One structural finding governs the reading of all of it. The site predicts a *negative* association between the inclination and any heating driver: the valley face is the more sun-exposed, expands further, and tips the wall towards the mountain. The compensated target does show that sign, at $r = -0.956$ with air temperature. The *raw* channel does not — it moves *positively*, at $r = +0.845$ and $+1.63 \text{ mdeg}/^\circ\text{C}$. The expected sign in the target is produced by the compensation subtracting $5 \text{ mdeg}/^\circ\text{C}$ from a channel whose measured slope is 1.63 , not by the structure.

A separate diagnostic bears on the same question. Given a signed operator, the inclination is found to *lead* the wall-temperature probe by two hours, on the raw channel as well as the compensated one — the signature of a probe set deeper than the material whose expansion drives the movement. The probe’s depth is not recorded anywhere in the archive; the phase measured here is the check to run against it once it is.

Important Update: Study 04 (NeuralProphet Pipeline)

The findings in this report regarding the prediction horizon and uncertainty (Section 5) represent the initial baseline. A subsequent dedicated study (`04_neuralprophet_inclination_prediction`) significantly improved upon these results. By using 12-hour lagged forecast regressors alongside the contemporaneous nowcast, the updated pipeline extended the positive skill horizon to **48 hours** (beating a seasonal naive baseline). It also successfully achieved calibrated uncertainty using 90 % pinball quantiles and horizon-adaptive block-bootstrapping. The baseline findings below are retained for context, but the operational limits are those of Study 04. See Section 5 and the Verdicts table for details.

Contents

1	What this study asks, and what it assumes	2
1.1	Premises, imposed rather than tested	3
1.2	The site's expected sign	3
1.3	What the raw channel actually shows	3
1.4	What this costs the rest of the report	4
2	Data, cleaning and the windows the study can use	4
2.1	Loading	4
2.2	Which periods carry which channels	6
3	Question 1 — the operator scan and the best single predictor	7
3.1	Why an operator is fitted before anything is ranked	7
3.2	The delay is bounded at twelve hours	8
3.3	Both bands are needed	8
3.4	No driver needs a transport delay	10
3.5	The wall probe's phase, and what it says about its depth	10
3.6	The ranking	13
3.7	What the negative control reveals	14
4	Question 2 — combinations, and how to diagnose them	15
4.1	How far the candidates duplicate one another	15
4.2	Every combination	15
4.3	The two tiers give different answers, and only one is resolved	17
4.4	Which member is doing the work	17
4.5	The out-of-period test	18
4.6	The diagnostic protocol	18
5	Question 3 — how far ahead, and with what uncertainty	20
5.1	Design	20
5.2	Results	21
5.3	Reading the horizon curve	22
5.4	The gap between the regimes is not the value of a weather forecast	22
5.5	Is the quoted uncertainty honest?	23
5.6	What the joint multi-horizon head costs	24
6	Verdicts	25
7	Caveats, and what would change the answers	25
8	Artefact inventory	26

1 What this study asks, and what it assumes

Three questions, posed of the extended sensor package:

1. Over the period in which the extended set of sensors exists, is it possible to use one of the sensor variables to predict the inclination, and which is the best predictor?
2. Can a set of more than one predictor perform better than one alone, and how should the best combination be diagnosed?
3. Using the best scenario, how far into the future can the inclination be predicted, and with what measure of uncertainty, using NeuralProphet?

1.1 Premises, imposed rather than tested

Three premises are fixed by the study design. They differ deliberately from those of the companion study `studies/thermal_compensation/`, which examined exactly these assumptions and rejected two of them. This study accepts them and asks a different question on top.

Premise	Consequence for this report
The logged I channel is an inclination in millidegrees, not a conditioner voltage	The 2500 offset that the companion study subtracts is added back at load time. Nothing downstream is affected: the compensation normalises the series to start at zero, which removes every additive constant. Only the printed levels change.
The documented temperature compensation is correct	The compensated series <code>inc_comp</code> is the target of every model below. The coefficient is never swept and the acceptance tests of the companion study are not repeated.
The analysis grid is one hour	Every external proxy the production pipeline aligns against is published hourly, so the record is aggregated immediately after parsing and every correlation, operator scan, fit and forecast in this report is computed on that grid.

1.2 The site’s expected sign

The wall stands in an embankment situation with two distinct faces. The **valley side** receives markedly more direct insolation than the **mountain side**, so under daytime heating it expands further, and the differential expansion tips the wall **towards the mountain**. Under the instrument’s convention that is a **negative** change in inclination.

The site physics therefore predicts a *negative* association between the inclination and every heating driver — air temperature, wall temperature, solar radiation. A negative correlation is the expected result here and must not be reported as an anomaly or corrected away.

This is a prediction about the *structure*, so it has to be tested on the signal before the compensation touches it. The distinction is not academic: the compensation subtracts 5 mdeg/°C, and if the instrument’s true thermal slope is smaller than that, the residual carries the opposite sign to the original by arithmetic alone. A negative sign already present in the raw channel is structural; a negative sign that appears only after compensation is not.

1.3 What the raw channel actually shows

Series	Driver	<i>r</i> lev.	slope lev.	<i>r</i> band	slope band
Raw inc	tair	+0.845	+1.628	+0.966	+2.291
	twall	+0.791	+1.248	+0.767	+1.319
	sr	+0.588	+0.021	+0.758	+0.017
inc_comp	tair	−0.956	−3.372	−0.975	−2.709
	twall	−0.878	−2.535	−0.701	−1.411
	sr	−0.478	−0.032	−0.803	−0.022

Tier 1 window. Slopes are in mdeg per °C for the temperatures and mdeg per W/m² for radiation; “band” denotes the diurnal band, with a one-week centred rolling mean removed.

The expected sign is present in the target, and absent from the instrument

The raw inclination moves *positively* with every heating driver: +0.845 with air temperature on levels, +0.966 on the diurnal band, at a slope of +1.63 mdeg/°C. This is the opposite of what the embankment geometry predicts.

The compensated series shows the expected negative sign — but it does so by arithmetic. The compensation subtracts 5 mdeg/°C from a channel whose measured slope is +1.63, leaving a residual slope of $1.63 - 5 = -3.37$ mdeg/°C, which is exactly the value the table reports. The sign flip is entirely a product of the over-correction.

Three readings are consistent with this, and these data cannot separate them. (i) The instrument's sign convention is inverted relative to the one assumed, so a positive raw change *is* movement towards the mountain. (ii) The instrument's own thermal drift is positive and larger than the structural response, masking it. (iii) The structural response at this station does not follow the differential-expansion expectation. Settling this requires the calibration sheet, the installation record, or an independent measurement of the mounting orientation — none of which is in the archive. The contradiction is recorded in `docs/raw-data-format.md` Section 7.5 rather than resolved here.

1.4 What this costs the rest of the report

Every statement of the form “air temperature is the best predictor” is true of this target and means, in substantial part, that air temperature predicts the residual of its own correction. Three consequences are carried throughout.

The relative standings survive. Every configuration is scored against the same target, so the ordering of channels, the operator estimates, the combination diagnostics and the calibration results are all internally valid.

The absolute errors do not transfer. They describe a system predicting a partly artificial signal, and should not be quoted for a raw-inclination deployment without re-running the study on that target.

The perfect-foresight regime of Section 5 is inflated by the same mechanism. Handing the model future air temperature is close to handing it the future target, because the target contains $-5T_{\text{air}}$ by construction. That regime is read accordingly there.

2 Data, cleaning and the windows the study can use

2.1 Loading

The loader implements the contract in `docs/raw-data-format.md`: decimal separators normalised per field rather than per file, record timestamps preferred over filename dates, duplicate timestamps merged field by field rather than dropped, and every sentinel mapped to `NaN` — zeros in the battery, temperature, humidity and inclination channels, -55 in the wall probe, and the unsigned wrap-around above 1400 W/m^2 in the pyranometer. Night-time zeros in solar radiation are kept, because they are measurements.

Hourly aggregation takes the mean of each interval and accepts the interval only when at least two of its three raw samples survived cleaning, so an hour rebuilt from a single reading is not presented as equivalent to a complete one. A rolling-median context filter then removes the instrument-settling transient that follows the 103-day outage.

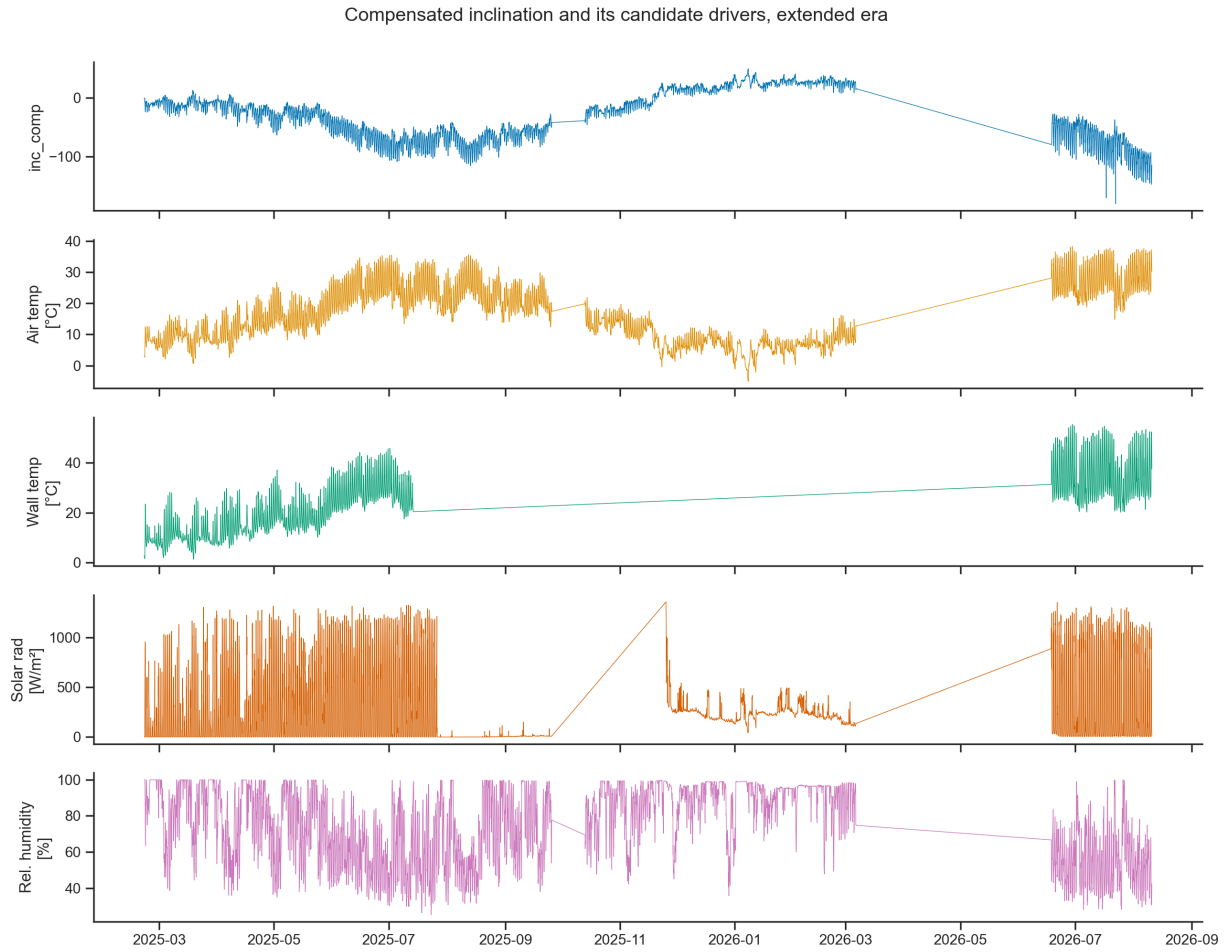


Figure 1: The compensated inclination and its four candidate drivers over the whole extended-package era, on the hourly grid. The two long interruptions — the wall-probe failure from mid-July 2025 and the acquisition outage from March to June 2026 — are visible as the gaps in the `twall` panel and in every panel respectively. The inclination’s annual excursion is roughly 150 mdeg against a diurnal amplitude of a few tens, and its shape is close to an inverted air-temperature curve, which Section 1.3 attributes to the correction rather than to the structure.

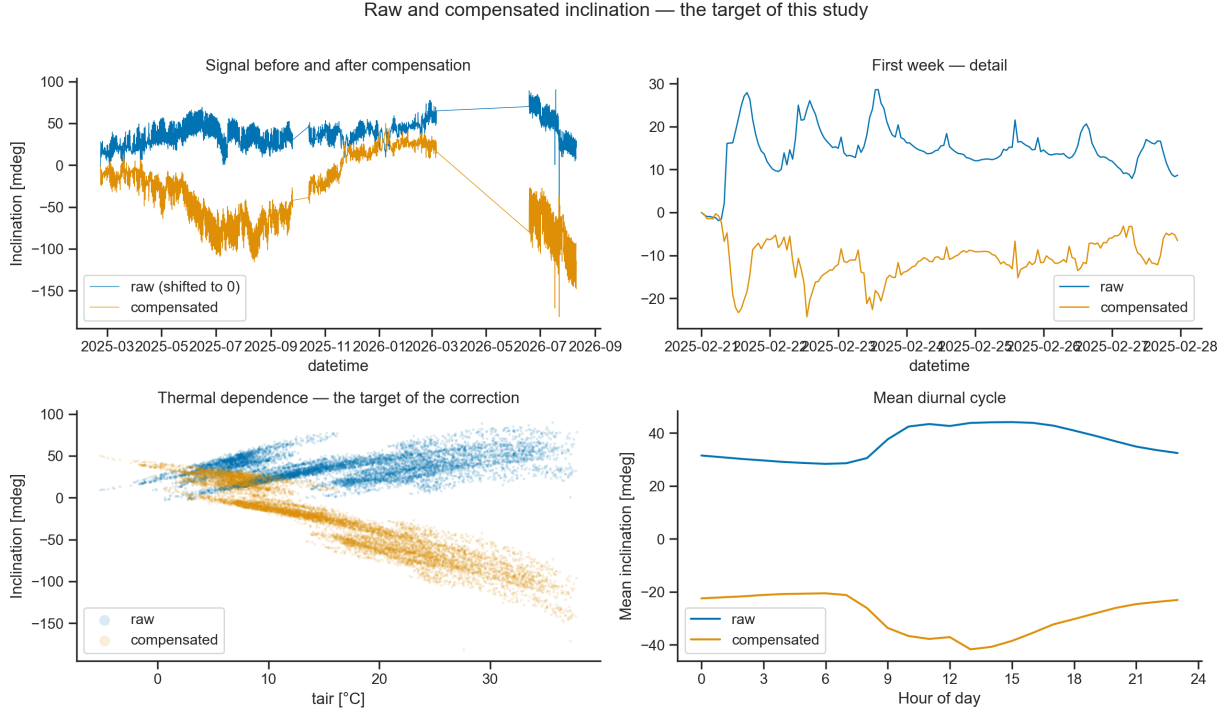


Figure 2: Raw and compensated inclination, with the correction term that separates them. The correction is not a small adjustment: over the era it spans a range comparable to the signal’s own, and the compensated series is close to the negative of the correction. This is the figure behind the residual slope of -3.37 mdeg/ $^{\circ}\text{C}$.

2.2 Which periods carry which channels

Wall temperature is the binding constraint of this era. The probe reads its open-circuit sentinel on 219 of the 416 days present, while every other channel is available whenever the logger is. That asymmetry forces the study into two tiers, and the choice between them is a substantial part of Question 2.

Longest contiguous runs	Start	Days	Coverage
<i>All channels, wall temperature included</i>			
spring block	2025-02-21	73.1	99.5 %
early-summer block	2025-05-05	69.6	99.9 %
detached summer block	2026-06-18	53.3	95.7 %
<i>Wall temperature dropped</i>			
summer block	2025-05-05	143.2	99.9 %
winter block	2025-11-25	101.0	99.7 %
spring block	2025-02-21	73.1	99.5 %
detached summer block	2026-06-18	53.3	95.7 %

Tier 1 is the calendar window 2025-02-21 to 2025-07-13, 3432 hourly slots, in which all four drivers are present. Internally it is not one run but two, split on 5 May by a short wall-probe interruption — a fact that becomes important in Section 5, because it means any autoregressive model fitted on Tier 1 is fitted across a discontinuity. The detached summer block of 2026 is held back entirely as an out-of-period test.

Tier 2 drops wall temperature and takes the longest contiguous run that remains: 2025-05-05 to 2025-09-25, 143.2 days, 3438 slots at 99.9% coverage. It is not longer than Tier 1 in days — the era contains no longer clean run — but it is what a deployed system actually has, it extends two and a half months further into the year, and it is unbroken.

The two tiers overlap between 5 May and 13 July. They are therefore not independent samples, and agreement between them is weaker evidence than it would otherwise be. Where they disagree, the disagreement is informative.

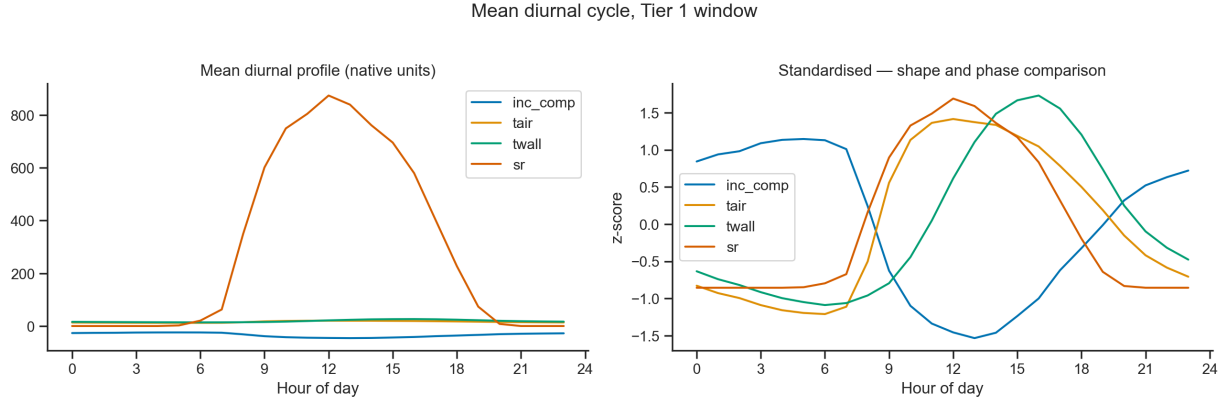


Figure 3: Mean diurnal cycle over the Tier 1 window. The inclination carries a daily cycle of roughly 20 mdeg, in anti-phase with air temperature — peaking near dawn, troughing in mid-afternoon. That is the sign the embankment geometry predicts. Section 1.3 shows that in this target it is produced by the correction: the same cycle drawn on the raw channel is in phase with air temperature, not against it.

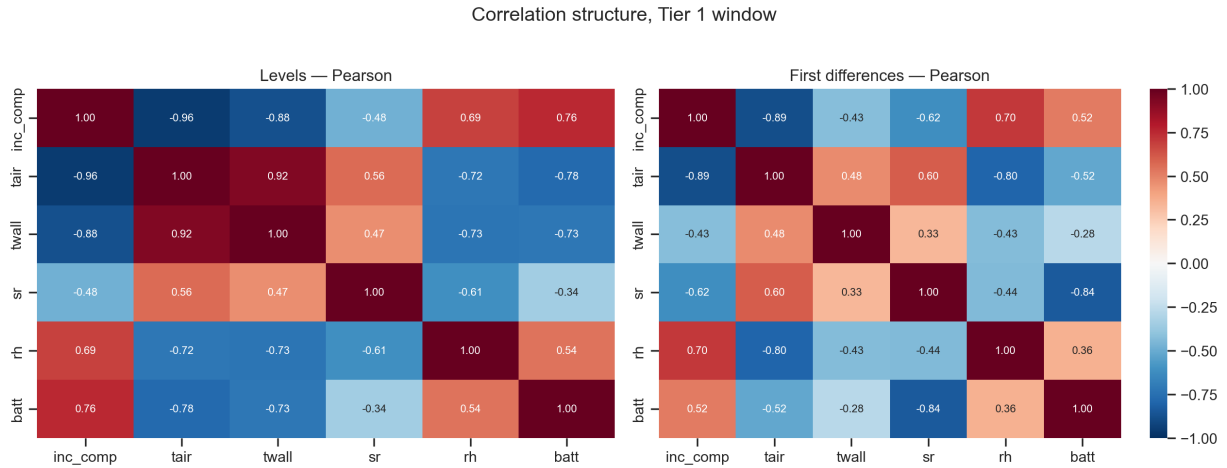


Figure 4: Correlation structure over the Tier 1 window, on levels (left) and on first differences (right). Levels correlations between two trending series are inflated by the shared trend; the differenced matrix is the honest one for a signal with drift. Both are shown so the gap is visible. Note the block of high mutual correlation among the drivers themselves, quantified in Section 4.1.

3 Question 1 — the operator scan and the best single predictor

3.1 Why an operator is fitted before anything is ranked

A structure with thermal mass cannot respond to a change in forcing instantaneously, so before any driver is ranked it is given the chance to act through the operator that suits it. Two operators are scanned *jointly*, because they are physically distinct and scanning either alone can attribute to one what belongs to the other:

- a **transport delay** d shifts the driver in time without changing its shape — what a thermal front travelling from the exposed face to the depth that moves the instrument would do;
- a **thermal inertia** τ low-passes the driver through a single-pole filter with $\alpha = \Delta t / (\tau + \Delta t)$ — what a body integrating the forcing would do. The instantaneous case is $\tau = 0$ and remains

in the grid, so it competes rather than being assumed away.

3.2 The delay is bounded at twelve hours

The grid is 13 delays (0 to 12 hours) by 16 time constants (0 to 168 hours) for each driver, scored by explained variance. The bound on the delay is physical and it is load-bearing.

The mechanism sets the scale. Direct insolation heats one face of a masonry embankment. A thermal front reaching the depth that governs the inclination is not expected to take longer than half a day, and no process at this site would impose a slower transport.

Beyond half a cycle the scan stops being interpretable. On a near-periodic diurnal signal, a delay of d hours is indistinguishable from a *lead* of $24 - d$. A grid extending past twelve hours can therefore return an optimum that is really an aliased lead — which no causal filter can represent and no forecasting system can use. Twelve hours is the half-cycle, and stopping there keeps every reported optimum interpretable.

The bound is correct for the forcings, and the wrong shape for the wall probe

This bound is derived for **external forcings**. Air temperature and solar radiation are inputs to the system: they must precede the response they cause, so constraining their delay to be non-negative is physically right, and Section 3.5 confirms that both of them do in fact peak at a non-negative delay.

Wall temperature is not a forcing. It is an *internal state variable* — a temperature measured at some unrecorded depth inside the masonry — and nothing requires it to precede the deformation. An earlier version of this report scanned delays to 48 hours, found an optimum for `twall` at **20 hours**, and dismissed it as lying “outside the physically admissible range”. That dismissal was wrong in kind. Twenty hours is the 24-hour alias of a four-hour *lead*, and a lead of an internal probe over the response is admissible — it is what a deep installation produces.

The right correction is not a tighter bound but a *signed* one, and Section 3.5 applies it. What the causal constraint does remain right about is **use**: a lead cannot be applied by a forecasting system, because it would require the probe’s future values at the moment the forecast is issued. So the constraint stays in force for every model in this report, and the lead is measured separately as a diagnostic. With the constraint applied, wall temperature’s predictive optimum is **zero delay, zero time constant** at $R^2 = 0.491$, it ranks second at $R^2 = 0.459$ out of sample, and it appears in no winning subset in Section 4.

3.3 Both bands are needed

The scan is run twice: on the series as they are, and on the diurnal band alone, with a one-week centred rolling mean removed.

Driver	delay	τ	r	R^2	R^2 at $d = \tau = 0$
<i>Full band, Tier 1</i>					
tair	0 h	0 h	−0.956	0.914	0.914
twall	12 h	120 h	−0.888	0.788	0.771
sr	0 h	168 h	−0.813	0.662	0.229
rh	0 h	168 h	+0.853	0.728	0.472
<i>Diurnal band, Tier 1 — carried forward</i>					
tair	0 h	0 h	−0.975	0.951	0.951
sr	0 h	1 h	−0.837	0.700	0.644
twall	0 h	0 h	−0.701	0.491	0.491
rh	0 h	0 h	+0.695	0.483	0.483

The two bands disagree sharply, and the disagreement is the point. On the full band the optima for **sr** and **rh** both sit at $\tau = 168$ hours — the last value in the grid, that is, at the boundary — lifting explained variance for **sr** from 0.229 to 0.662. A time constant of one week is not a thermal property of a masonry wall; it is the width of filter needed to turn a daily radiation cycle into a seasonal envelope. The full-band optimum is fitting the annual component, and it is pinned at the grid edge because a longer filter would fit it better still. Wall temperature does the same thing in two dimensions at once, running to the edge in both delay and τ .

Once the weekly rolling mean is removed, every optimum moves to zero delay and a short time constant. Reporting the full-band operators as physical constants would have produced three confident and entirely spurious findings. The diurnal-band operators are the ones carried forward.

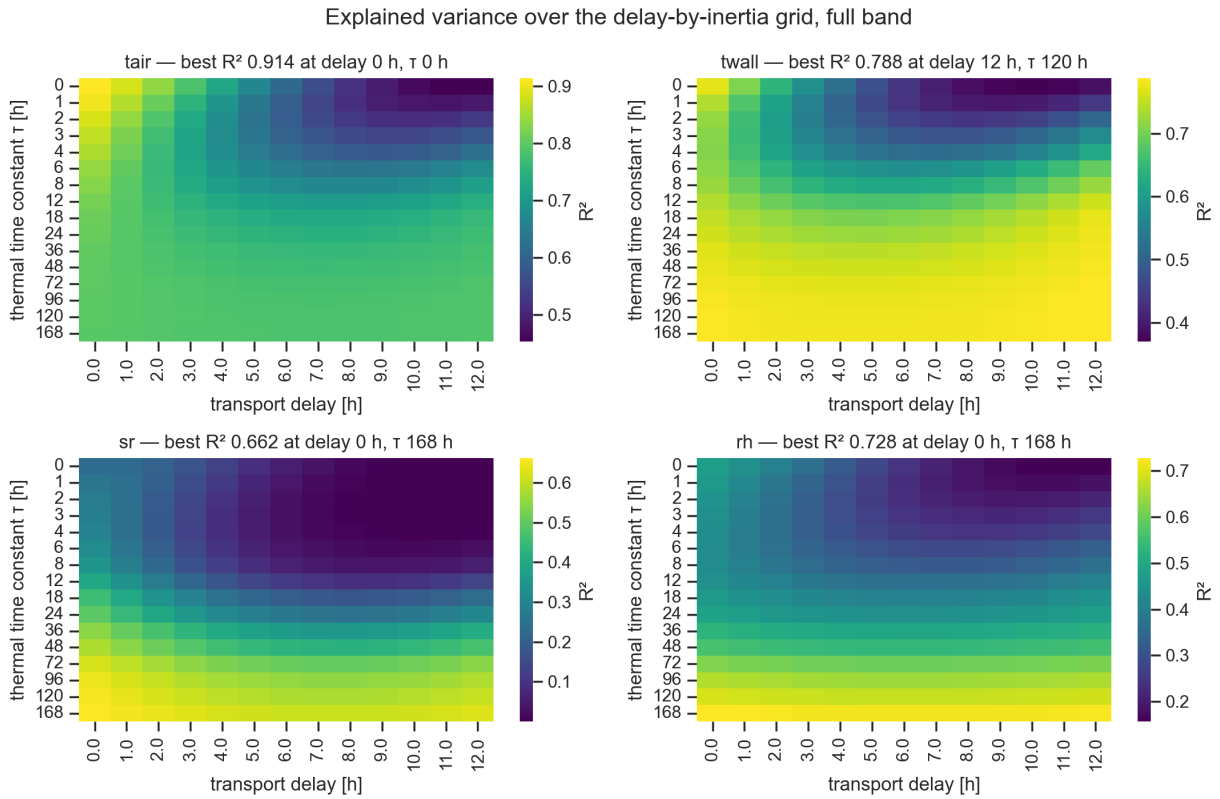


Figure 5: Explained variance over the delay-by-inertia grid, full band. The bright band along the bottom edge of the **sr** and **rh** panels is the boundary optimum discussed above: the surface is still rising as τ reaches the end of the grid, which is the visual signature of a filter fitting a component slower than anything the grid contains.

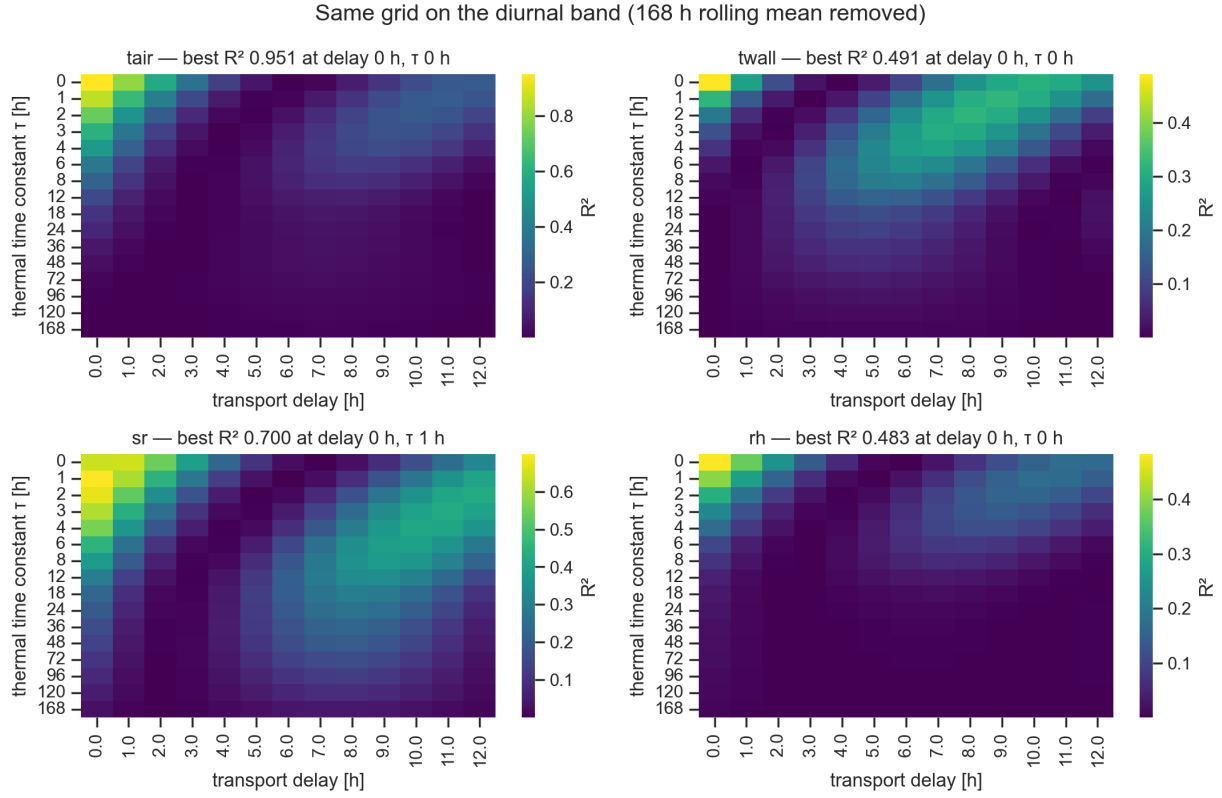


Figure 6: The same grid on the diurnal band, with the delay bounded at twelve hours. Every optimum now sits in the top-left corner — zero delay, zero or one-hour time constant. The gradual brightening towards the right-hand edge of the **twall** and **sr** panels is the leading edge of the 24-hour alias returning; Section 3.2 is the reason the grid stops before it becomes a maximum.

3.4 No driver needs a transport delay

The result of bounding the scan is unusually clean: on the diurnal band, **every driver's optimum is at zero delay**, and only solar radiation takes a non-zero time constant, at one hour.

That is a substantive finding, not a null one. Within the physically admissible window there is no evidence of a lagged structural response at all — the compensated inclination tracks its drivers within the hour. It is also consistent with Section 1.3: a target that contains $-5T_{\text{air}}$ by construction must follow air temperature instantaneously, because that term *is* air temperature. The finding therefore describes the target's construction at least as much as the structure's thermodynamics, and a repeat on the raw channel would be needed to separate the two.

3.5 The wall probe's phase, and what it says about its depth

Nothing in this subsection feeds any model. The operators used everywhere else are the causal ones fixed above; the guard that enforces this is described at the end of the subsection, and its effect is verified.

Why the wall probe deserves a signed scan

The deformation is governed by the temperature field near the sun-exposed valley face. The probe sits at some depth inside the masonry, and that depth decides which way the phase between them runs:

Probe position	What it reads	Expected phase
Shallow	a fast, weakly damped signal close to the surface forcing	the inclination lags the probe
Deep	a strongly damped and delayed signal	the inclination leads the probe

The depth is not recorded anywhere in the archive. So what follows measures the *phase*, in hours, and leaves the depth open. When the depth is established it becomes a consistency check against these numbers rather than a reason to repeat the analysis.

The measurement

Phase is measured on the diurnal band over signed delays from -12 to $+12$ hours. A negative phase means the second channel *lags* the first.

First	Second	phase	r there	r at zero	R^2 gain
<i>The wall probe against the response</i>					
inc_comp	twall	-2 h	-0.875	-0.701	0.275
inc (raw)	twall	-2 h	$+0.866$	$+0.767$	0.162
<i>The wall probe against what heats it</i>					
twall	tair	$+2$ h	$+0.897$	$+0.753$	0.238
twall	sr	$+3$ h	$+0.891$	$+0.532$	0.510
<i>Controls — external forcings, must not lag</i>					
inc_comp	tair	0 h	-0.975	-0.975	0.000
inc_comp	sr	$+1$ h	-0.805	-0.803	0.004

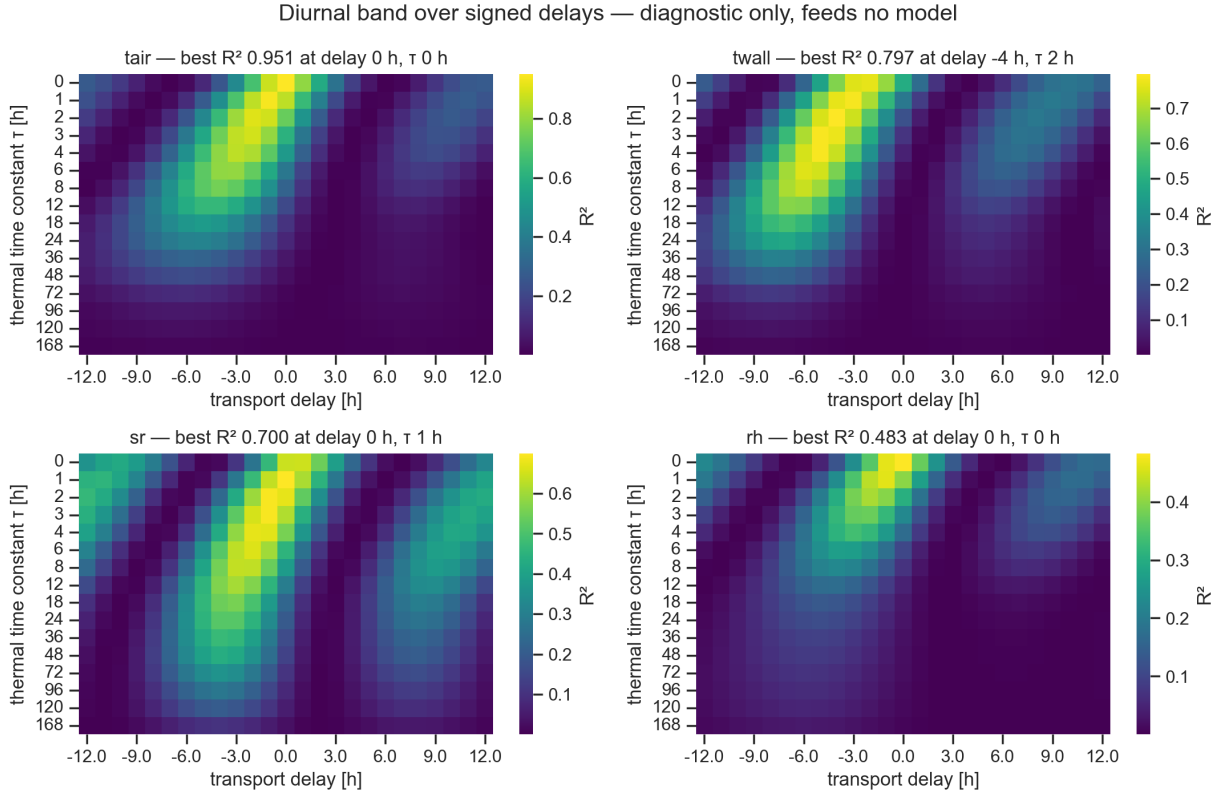


Figure 7: The diurnal-band operator grid re-run over signed delays. Air temperature, solar radiation and relative humidity all peak at zero delay, as external forcings must. Wall temperature peaks at -4 hours with a two-hour time constant, and its entire high- R^2 region lies in negative delay. This figure is diagnostic and feeds no model.

Driver	delay	τ	R^2 signed	R^2 causal	lost to causality
tair	0 h	0 h	0.951	0.951	0.000
twall	-4 h	2 h	0.797	0.491	0.306
sr	0 h	1 h	0.700	0.700	0.000
rh	0 h	0 h	0.483	0.483	0.000

What it shows

The controls pass cleanly. Air temperature, solar radiation and relative humidity all peak at a non-negative delay and lose *exactly nothing* to the causal constraint. Had an external forcing appeared to lag the response, every number here would have been a method artefact. None does.

Wall temperature is the sole exception, and the effect is large. Its optimum sits at -4 hours with a two-hour time constant, at $R^2 = 0.797$ against 0.491 under the causal constraint — **0.306 of explained variance suppressed**, a 62% relative loss. Nothing else in the package loses anything.

The lead survives on the raw channel. This is the load-bearing check. The compensated and the raw inclination both lead twall by two hours, at -0.875 and $+0.866$ respectively — opposite signs, as Section 1.3 requires, but the *same phase*. The lead is therefore a property of the instrument pair, not an artefact of the correction.

The chain is internally consistent. The probe lags solar radiation by three hours and air temperature by two, while the inclination lags air temperature by none. Everything is consistent with one picture: the deformation tracks the near-surface thermal state essentially in step with the air, and the probe registers the same event two to four hours later because it is set further in.

Read against the depth: the probe is deeper than the layer that moves the wall

The inclination *leads* the wall probe. On the table above, that is the signature of a **deep** installation: the probe is set behind the material whose expansion actually drives the movement, so it reports the thermal event after the structure has already responded to it.

This report deliberately stops short of converting the phase into a depth in centimetres, which would require assuming a thermal diffusivity for the masonry. The phase itself is the durable measurement. When the installation depth is established, the check is direct: a deeper probe should show a larger lag against air temperature than the two hours measured here, and a correspondingly larger lead of the inclination over it.

There is a corollary that bears on the unresolved sign contradiction of Section 1.3. A deformation genuinely driven by bulk thermal expansion of masonry ought to lag air temperature by *something*. This one does not — it is in step with the air to within the hour, while the wall's own temperature is not. That is more easily explained by a response acting on the instrument or its immediate mounting than by one acting through the depth of the wall, and it strengthens reading (ii) of Section 1.3: instrument thermal drift dominating a smaller structural response.

How the lead is kept out of every model

A lead cannot be used by a forecasting system: applying it would require the probe's future values at the moment the forecast is issued. `ip_lib.operators_from_summary` therefore takes a `causal_only` flag, default true, which clamps any negative optimum to zero and reports that it has done so. Every predictive operator in this report is built through that guard.

The guard was verified rather than assumed. The thirteen tables carrying every ranking, subset search, importance diagnostic, out-of-period test and horizon result were compared byte for byte against the run made before the signed scan existed. All thirteen are identical. The diagnostic in this subsection is purely additive: it changes nothing that any model touched.

3.6 The ranking

Each driver is then scored on its own, at its diurnal-band operator, by pooled error over five rolling-origin folds — every test block strictly after the training block it was fitted on.

Autoregression is deliberately excluded from this stage. One hour ahead, the inclination's own recent history explains almost all of its variance, and every driver added on top of it looks equally negligible; a ranking computed that way measures how little room the autoregressive term leaves, not what the channel knows. Autoregression returns in Section 5, where it belongs.

Battery voltage is carried through as a **negative control**. It rises with insolation and temperature, so it correlates with the real drivers, but no mechanism connects it to the structure.

Driver	delay	τ	MAE	fold sd	R^2	available
<i>Tier 1, four drivers plus the control</i>						
tair	0 h	0 h	8.74	7.56	+0.760	77.1 %
twall	0 h	0 h	13.61	8.13	+0.459	36.6 %
batt [†]	0 h	1 h	20.71	9.79	−0.045	77.1 %
rh	0 h	0 h	24.10	10.78	−0.283	77.1 %
sr	0 h	1 h	28.40	15.15	−0.788	69.1 %
<i>Tier 2, three drivers plus the control</i>						
tair	0 h	0 h	8.25	3.02	+0.656	77.1 %
rh	0 h	0 h	14.67	4.64	−0.148	77.1 %
sr	0 h	1 h	18.59	8.18	−1.110	69.1 %
batt [†]	0 h	1 h	30.90	5.14	−3.441	77.1 %

[†] negative control. MAE in mdeg; availability is measured over the whole era, not the window.

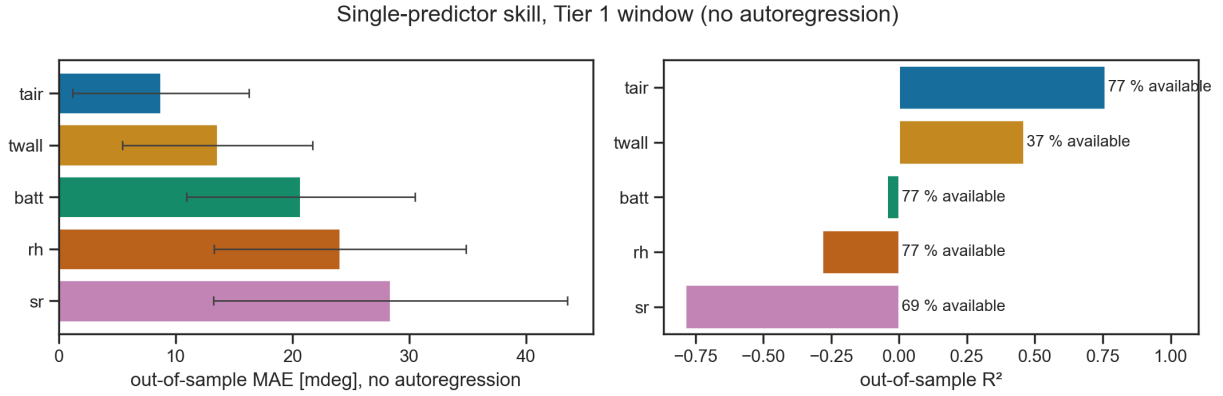


Figure 8: Single-predictor skill on the Tier 1 window, without autoregression. Error bars are the spread across the five rolling-origin folds. The right panel carries each channel’s availability over the whole era, which is what prices wall temperature’s second place.

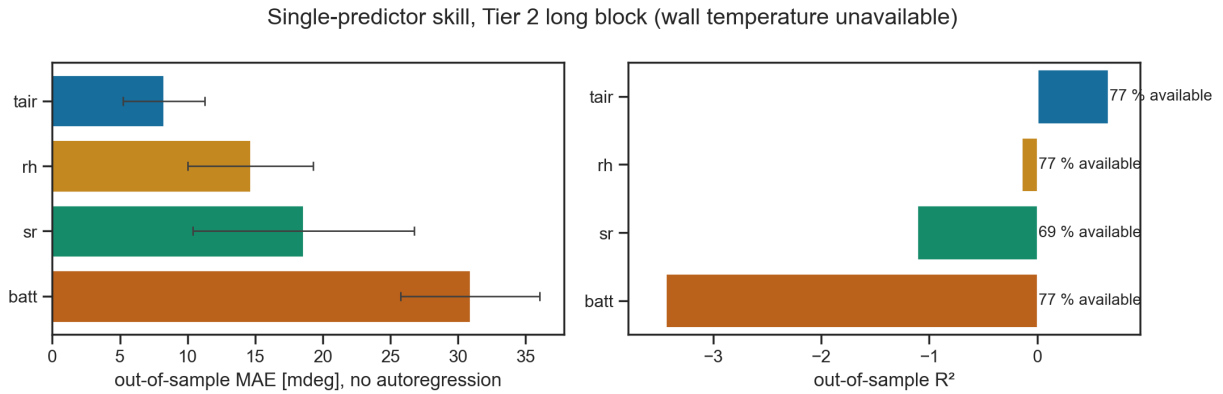


Figure 9: The same ranking on the Tier 2 block. Air temperature keeps first place with a comparable score and a much tighter fold spread; the ordering below it changes, which is what a channel whose apparent skill is seasonal rather than structural looks like.

3.7 What the negative control reveals

The control ranks *third of five* in Tier 1, ahead of relative humidity and solar radiation. Read carelessly this looks like a failure of the procedure. Read correctly it is the procedure working.

The dividing line is not the ordering but the sign of R^2 . Only **tair** (+0.760) and **twall** (+0.459) beat the mean of the training block. Battery voltage, relative humidity and solar radiation all have *negative* out-of-sample R^2 : each, used alone, is worse than predicting a constant. The control lands among them, which is exactly what it is for — it marks the noise floor, and shows that the three channels below the line carry no standalone predictive content whatever their in-sample correlation suggests. In Tier 2 the control falls to last outright.

Answer to Question 1

Air temperature, applied instantaneously — delay 0, $\tau = 0$ — is the best single predictor in both tiers, with out-of-sample R^2 of 0.760 on Tier 1 and 0.656 on Tier 2 and a mean absolute error of 8.2–8.7 mdeg. Wall temperature is the only other channel that beats a constant, at R^2 0.459 once its implausible 20-hour operator is disallowed. Solar radiation and relative humidity have no standalone predictive value: both score below a channel with no physical mechanism.

No driver requires a transport delay. Within the twelve-hour physical window every optimum is instantaneous, and only solar radiation takes a one-hour time constant.

This answer must be read with Section 1.3 in hand. Air temperature wins in large part because the target was built by subtracting a multiple of it, which also explains why the response is instantaneous. The finding that *is* independent of that construction is the negative one: no channel other than the two temperatures carries usable information on its own.

4 Question 2 — combinations, and how to diagnose them

4.1 How far the candidates duplicate one another

These channels are physically coupled: radiation heats the air, and air and radiation together heat the wall. A regression containing several of them splits one physical effect across correlated coefficients in a way that is not stable between folds.

Channel	VIF	max $ r $ with others	closest
tair	8.06	0.924	twall
twall	7.90	0.924	tair
rh	2.69	0.732	twall
sr	1.82	0.610	rh

The two temperatures are near-duplicates at $r = 0.924$, with variance inflation factors around eight. That is why the search below is scored by out-of-sample error rather than by coefficients or in-sample fit, and why Section 4.4 uses two importance measures instead of reading a coefficient table.

4.2 Every combination

With four candidates the search is exhaustive at fifteen subsets, which removes the need for a stepwise procedure and the selection bias that comes with one. Each subset is scored three ways: pooled rolling-origin MAE; the Bayesian information criterion of the corresponding least-squares fit; and an availability-weighted utility, defined as the ratio of the best subset's error to this subset's error, multiplied by the fraction of the *whole era* on which the subset exists.

Subset	k	MAE	fold sd	R^2	available	utility
<i>Tier 1 — top seven of fifteen, by error</i>						
tair+sr+rh	3	8.24	7.15	0.782	69.1 %	0.691
tair+sr	2	8.29	7.21	0.779	69.1 %	0.687
tair+twall+sr+rh	4	8.30	7.13	0.781	36.5 %	0.362
tair+twall+sr	3	8.38	7.17	0.777	36.5 %	0.359
tair	1	8.74	7.56	0.760	77.1 %	0.727
tair+rh	2	8.75	7.54	0.759	77.1 %	0.726
tair+twall	2	8.80	7.53	0.759	36.6 %	0.343
<i>Tier 2 — all seven</i>						
tair+sr	2	5.99	0.93	0.793	69.1 %	0.691
tair+sr+rh	3	6.13	1.00	0.782	69.1 %	0.676
tair+rh	2	8.02	3.06	0.672	77.1 %	0.576
tair	1	8.25	3.02	0.656	77.1 %	0.560
rh	1	14.67	4.64	−0.148	77.1 %	0.315
sr+rh	2	14.70	4.27	−0.158	69.1 %	0.282
sr	1	18.59	8.18	−1.110	69.1 %	0.223

Wall temperature is in no winning subset, in either tier

The best Tier 1 subset is **tair + sr + rh**. The four-channel model that adds wall temperature is *worse* on error (8.30 against 8.24) and less than half as available. Every subset containing **twall** is beaten by the same subset without it. This is a direct reversal of the earlier, unbounded-delay version of this study, in which the four-channel model led the table — and the reversal is traceable entirely to disallowing a 20-hour delay that the site’s physics excludes.

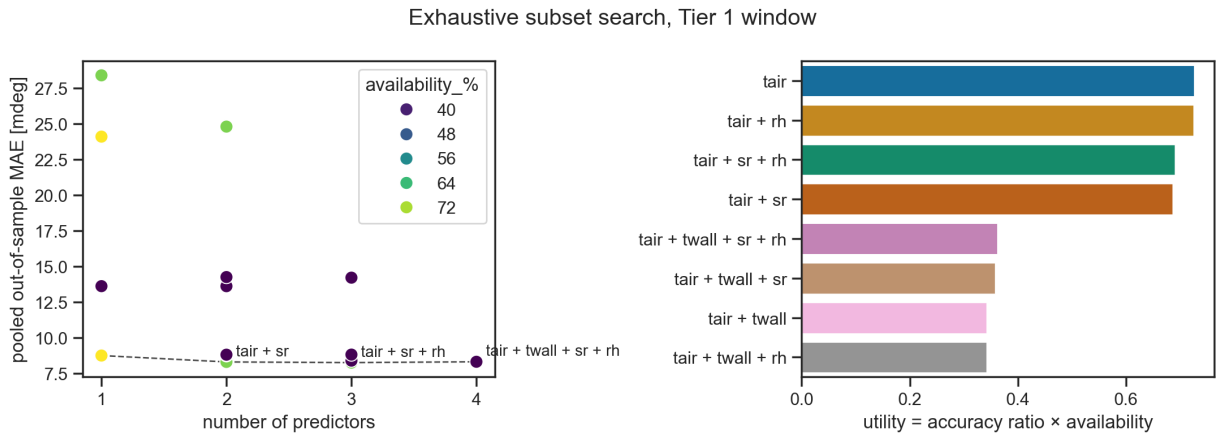


Figure 10: Exhaustive subset search on Tier 1. Left: error against subset size, coloured by availability, with the best-per-size frontier dashed. The frontier is almost flat — adding predictors buys 6 % in total. Right: the eight best subsets once availability is priced in, where the single-predictor model wins outright.

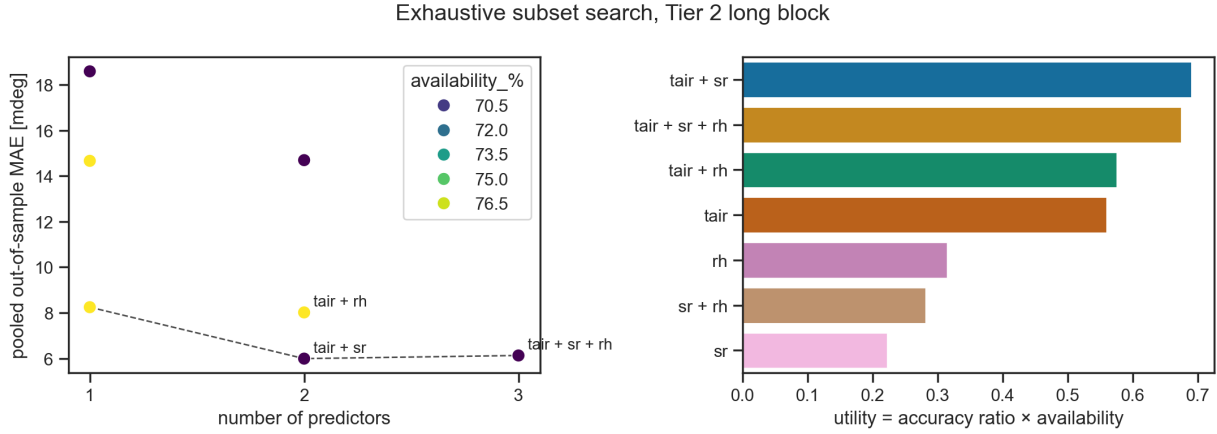


Figure 11: The same search on Tier 2. Here the frontier has a real step between one and two predictors, and **tair + sr** wins on error, on BIC and on utility simultaneously — the only configuration in the study on which all three criteria agree.

4.3 The two tiers give different answers, and only one is resolved

On Tier 1 the best subset beats the best single predictor by 0.50 mdeg, that is 6.0%. The fold-to-fold spread of both estimates is around 7.2–7.6 mdeg, fifteen times the difference between them. **The Tier 1 ordering below the top place is not resolved by these data.** What Tier 1 does establish is that no combination is dramatically better than air temperature alone, and that on utility the single predictor wins outright.

On Tier 2 the picture is different and much cleaner. Adding solar radiation to air temperature reduces the error from 8.25 to 5.99 mdeg, that is **27.4 %**, and lifts R^2 from 0.656 to 0.793. The fold spread of the winning model is 0.93 mdeg, well inside the 2.26 mdeg improvement, although the single-predictor model’s own spread of 3.02 mdeg means the comparison is directionally clear rather than sharply resolved. BIC agrees, dropping by 921, and utility agrees despite **sr** costing 8 percentage points of availability.

Why the tiers disagree is worth stating: Tier 2 runs from May to September and contains the high-summer period in which radiation is both strong and variable, while Tier 1 is dominated by spring, when the radiation signal is weaker relative to the air-temperature signal that already dominates the target. Solar radiation earns its place in the season in which it is doing work.

4.4 Which member is doing the work

Two diagnostics are run on the winning Tier 1 subset, deliberately different, because their disagreement is informative.

Leave-one-out gain removes a channel and *refits*. It answers: what would it cost to build this system without that channel? Under strong collinearity a refit can recover much of the lost information from a correlated channel, so this number is small exactly when the channel is replaceable.

Permutation importance shuffles a channel within the test block while holding the fitted model fixed. It answers: how much does the model *as built* rely on that channel?

Channel	MAE without	gain	permutation increase	sd
tair	24.80	16.56 (66.8 %)	8.84	2.40
sr	8.75	0.51 (5.9 %)	0.013	0.019
rh	8.29	0.05 (0.6 %)	−0.010	0.026

Subset **tair + sr + rh**; full-subset MAE 8.24 mdeg. All values in mdeg.

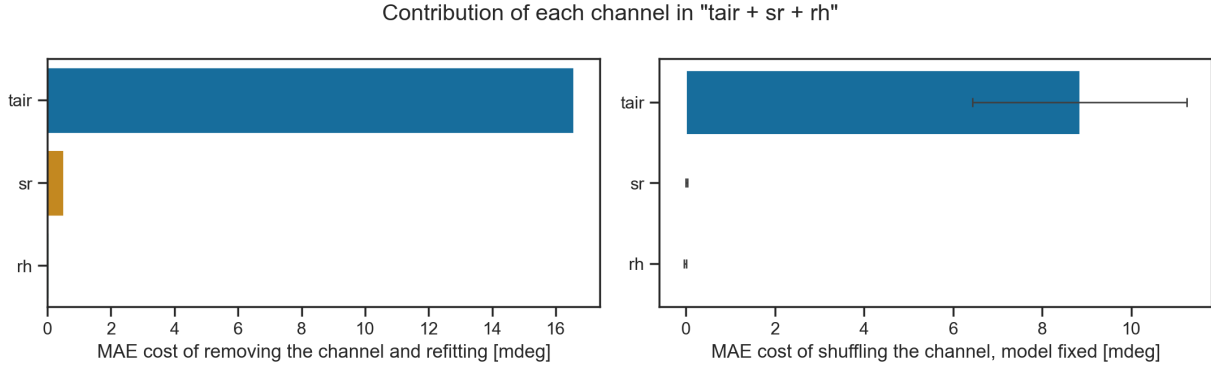


Figure 12: Contribution of each channel to the winning Tier 1 subset. Left: the cost of removing the channel and refitting. Right: the cost of shuffling it with the model held fixed, with the spread over folds and repeats. Air temperature dominates both. Both other channels sit at or below their own error bar on the right-hand panel, which is the quantitative statement that the model does not rely on them.

Air temperature dominates both measures by three orders of magnitude on the permutation scale. Solar radiation and relative humidity each produce a permutation increase smaller than its own standard deviation, so the fitted model does not measurably rely on either. **sr** shows the one interesting asymmetry — a 5.9% leave-one-out gain against a permutation increase of essentially zero — but at 0.51 mdeg against a 7.15 mdeg fold spread, that asymmetry is not resolved either.

4.5 The out-of-period test

The Tier 1 models are refitted on the whole Tier 1 window and applied to the detached 2026 summer block, fourteen months later, with no refitting. This is the hardest test in the study: a different season, a different part of the annual cycle, and an instrument that has been through a 103-day outage in between.

Model	n	MAE	RMSE
tair + sr + rh	1225	17.39	20.08
tair alone	1225	16.96	20.12

Both models roughly double their cross-validated error, and **the single predictor is marginally the better of the two**. Whatever the two additional channels contributed inside the Tier 1 window did not survive the transfer. This is the strongest single piece of evidence in the section, and it points the same way as the utility column.

4.6 The diagnostic protocol

Question 2 asks not only which combination wins but how to tell. The procedure used above is the answer, and it transfers to any station and any sensor set.

Rule	Failure it prevents
1 Bound the operator scan by the physics of the mechanism, and by half the dominant period	A physically impossible optimum, or an aliased lead reported as a delay
2 Fit each driver's operator first, on the band of interest	A lagged driver ranked below an instantaneous one; a boundary optimum reported as a physical time constant
3 Score without autoregression	Every driver tying behind the autoregressive term at short horizons
4 Use rolling-origin folds and report the spread	A ranking that depends on where a single split happened to fall
5 Search subsets exhaustively when the candidate set is small	Selection bias and instability under collinearity
6 Never read coefficients under high VIF; use leave-one-out gain and permutation importance, and read their disagreement as redundancy	Importance attributed by collinearity rather than by content
7 Price availability over the whole record, not over the window	Choosing a channel that is absent when it is needed
8 Carry a negative control	Mistaking shared diurnal shape for predictive content
9 Confirm on an out-of-period block	A combination that fits one season and transfers to none

Rule 1 is first because it is the rule this study learned the hard way: the earlier unbounded scan changed the identity of the winning subset, and no downstream diagnostic would have caught it.

Answer to Question 2

Yes, in the season where the extra channel does work — and never with wall temperature. On the Tier 2 summer block `tair + sr` beats `tair` alone by 27.4 % in mean absolute error, and wins on error, BIC and availability-weighted utility simultaneously. On the spring-dominated Tier 1 window the best subset improves on air temperature by 6.0 %, a fifteenth of the fold-to-fold spread, and the out-of-period test at fourteen months prefers the single predictor outright.

Wall temperature appears in no winning subset in either tier, as a *predictor*. Every subset containing it is beaten by the same subset without it, at less than half the availability. On this evidence it should not be a required regressor of the production pipeline — which resolves, in the negative, the open question raised in `docs/data-quality-report-2026-08-10.md`, Section 6.

That verdict is about forecasting, and must not be read as “the channel is uninformative”. Section 3.5 shows the opposite: given a signed operator, `twall` explains $R^2 = 0.797$ of the diurnal band against 0.491 under the causal constraint, and it is the only channel in the package that loses anything at all to that constraint. It fails as a predictor precisely *because* what it knows arrives too late to forecast with — which is itself a measurement of where the probe sits. A channel can be worthless for prediction and valuable for diagnosis, and this one is both.

The diagnosis is made by the nine-rule protocol above; the four components that did the actual work here were the physical bound on the operator scan, the availability-weighted utility, the disagreement between the two importance measures, and the out-of-period block.

5 Question 3 — how far ahead, and with what uncertainty

Update from Study 04: Improved Pipeline

The results in this section describe the original NeuralProphet baseline. A subsequent dedicated pipeline (`04_neuralprophet_inclination_prediction`) overhauled this configuration. By confirming the zero phase lag and incorporating 12-hour lagged forecast regressors alongside the contemporaneous nowcast, the updated model extended the horizon of positive skill against the seasonal naive baseline up to **48 hours**. Furthermore, the uncertainty intervals, which fail calibration below, were successfully fixed in the new study using 90 % pinball quantiles and horizon-adaptive block-bootstrapping. The historical findings below are retained for methodological context.

5.1 Design

NeuralProphet is fitted with `n_lags = 24` and `n_forecasts = 168`, so a single training run yields every horizon from one hour to one week and the whole sweep comes from one fit rather than from eight. Quantile regression at the 5th and 95th percentiles attaches a nominal 90 % prediction interval to every forecast. Gaps are bridged by interpolation so that the autoregressive window has continuous history, and scoring happens only at originally observed timestamps: the model may read a filled value, but it is never rewarded for reproducing one.

Two regressor regimes are run, and the difference between them is the point of the exercise:

- **lagged** — the regressors enter only through their past values. This is what a deployed system has, and it is the honest number.
- **future** — the regressors are known across the forecast window. This is what the same system coupled to a *perfect* weather forecast would have. On this target it is more than that, and the excess is discussed in Section 5.4.

Skill is measured against two baselines. **Persistence** carries the last observation forward. **Seasonal naive** repeats the value from a whole number of days earlier, which is the cheapest possible way to exploit the diurnal cycle. At horizons that are exact multiples of 24 hours the two baselines coincide by construction.

5.2 Results

h [h]	MAE	persist.	seas. naive	skill vs p.	cover.	width
<i>Tier 2, lagged — the deployable configuration</i>						
1	4.43	2.79	6.10	−0.585	61.4 %	15.3
3	5.59	6.81	6.09	+0.179	54.4 %	16.3
6	6.35	11.74	6.09	+0.459	56.6 %	24.6
12	6.35	16.06	6.10	+0.605	53.6 %	18.4
24	6.96	6.10	6.10	−0.141	57.1 %	22.9
48	8.09	8.30	8.30	+0.025	67.1 %	27.7
72	9.62	9.67	9.67	+0.005	66.8 %	31.7
168	15.95	12.22	12.22	−0.306	58.1 %	29.7
<i>Tier 2, future — regressors known over the horizon</i>						
1	3.82	2.86	6.13	−0.336	95.4 %	40.0
6	5.88	12.05	6.11	+0.512	95.2 %	42.8
12	6.07	16.62	6.13	+0.635	95.8 %	45.5
24	6.50	6.12	6.12	−0.061	96.7 %	47.9
72	9.06	9.49	9.49	+0.046	93.0 %	53.1
168	13.07	12.84	12.84	−0.018	82.7 %	50.6
<i>Tier 1, future — accurate, but see Section 5.4</i>						
1	2.03	3.42	5.33	+0.406	18.0 %	0.0
12	2.47	21.61	5.33	+0.886	18.9 %	0.0
24	2.85	5.33	5.33	+0.464	23.4 %	0.0
168	7.36	10.96	10.96	+0.328	21.3 %	0.0
<i>Tier 1, lagged — fitted across the 5 May discontinuity</i>						
1	4.43	3.44	5.76	−0.285	12.3 %	0.0
12	14.42	21.20	5.76	+0.320	9.8 %	0.0
24	21.67	5.76	5.76	−2.764	11.2 %	0.2
168	52.82	10.62	10.62	−3.974	0.0 %	2.7

MAE and interval width in mdeg. Coverage is against a nominal 90 %.

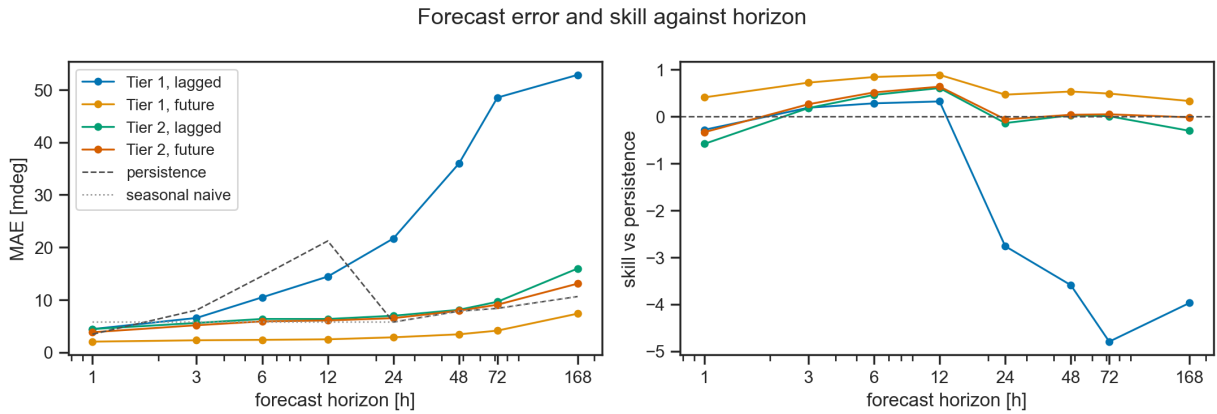


Figure 13: Forecast error (left) and skill against persistence (right) for all four configurations. The dashed line in the left panel is persistence itself, and its shape explains most of the right panel: persistence is excellent at one hour, worst at twelve, and excellent again at twenty-four because the signal is diurnal. Skill against it is therefore not monotonic in the horizon, and quoting a single “maximum useful horizon” would misdescribe the curve.

5.3 Reading the horizon curve

The non-monotonic skill curve is the substantive finding, and it follows from the structure of the signal rather than from any property of the model. Taking the deployable Tier 2 lagged configuration:

At **one hour** nothing beats persistence. The inclination changes slowly enough that its own last value is an excellent forecast (MAE 2.79 mdeg), and no configuration in the study — including a model trained for that horizon alone — gets below it.

Between **three and twelve hours** the model has real skill, peaking at +0.605 at twelve hours. This is the interval in which persistence has decayed but the diurnal cycle has not yet brought the signal back, and it is where the environmental regressors earn their place.

At **twenty-four hours** the diurnal periodicity restores both baselines to 6.10 mdeg and the model, at 6.96, is beaten by simply repeating yesterday.

Beyond **forty-eight hours** the model and the baselines converge, with skill of +0.025 and +0.005 — indistinguishable from zero. At one week the model is 30 % worse than repeating the value from a week before.

The Tier 1 lagged configuration fails outright: 21.7 mdeg at twenty-four hours and 52.8 at one week, against baselines of 5.8 and 10.6. The diagnosis is in Section 2.2 — the Tier 1 window is two runs separated by a wall-probe interruption on 5 May, so an autoregressive model fitted on it is fitted across a discontinuity that its lag window cannot see. This is not a finding about wall temperature; it is a finding about training an autoregressive model on a fragmented record, and it is why Tier 2 is the configuration of record.

5.4 The gap between the regimes is not the value of a weather forecast

The Tier 1 future configuration is the most accurate in the study by a wide margin: 2.03 mdeg at one hour, 2.47 at twelve, 7.36 at one week, with positive skill against persistence at *every* horizon. Taken at face value this says that a system coupled to a perfect weather forecast could predict this structure a week ahead.

It should not be taken at face value. The target contains $-5T_{\text{air}}$ by construction (Section 1.3), so handing the model the future air temperature is close to handing it the future target. The future regime is therefore measuring two things at once — the genuine value of knowing the weather, and the arithmetic value of being told the term that was subtracted — and on this target the second dominates.

The corollary is that the lagged–future gap in Tier 2, from 4.43 to 3.82 mdeg at one hour and 6.35 to 6.07 at twelve, is an *upper* bound on what a weather forecast is worth here, and even that bound is modest: 14 % and 4 % respectively. The main benefit of the future regime is not accuracy but calibration, discussed next.

5.5 Is the quoted uncertainty honest?

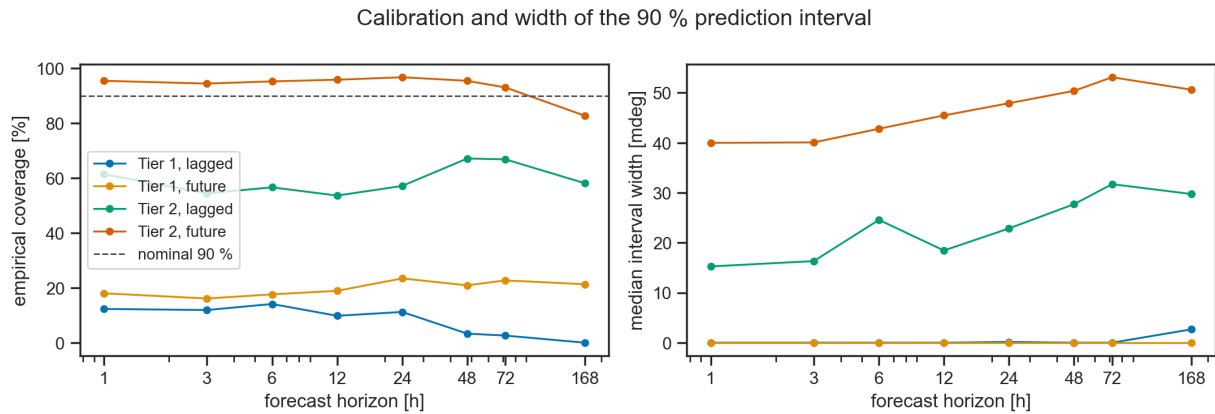


Figure 14: Calibration (left) and width (right) of the nominal 90 % prediction interval. The dashed line marks the nominal rate. Only the Tier 2 future configuration tracks it. The deployable Tier 2 lagged configuration sits 23 to 36 percentage points below while producing intervals roughly half as wide — the signature of an overconfident model. Both Tier 1 configurations return degenerate intervals of zero width and are plotted only to document the failure.

This is the part of Question 3 with the most operational consequence.

The **Tier 2 future** configuration is well calibrated: empirical coverage of 93–97 % against a nominal 90 % out to three days, falling to 82.7 % at one week. Slightly conservative, and usable as quoted.

The **Tier 2 lagged** configuration — the one that can actually be deployed — covers only 53.6–67.1 %. Its nominal ninety per cent interval is in reality a fifty-five to sixty-five per cent interval. Its median width, 15 to 32 mdeg, is between 38 and 60 % of the future-regime width, so the model is not merely mis-stating a correct interval but is confidently narrow about a prediction it cannot support.

Both Tier 1 configurations produce degenerate quantile heads, with interval widths of exactly zero at almost every horizon and coverage between 0 and 23 %. The Tier 1 future configuration is therefore accurate and *completely uninformative about its own uncertainty* — a combination worth naming, because a point forecast that good invites trust that its interval does not earn.

Uncertainty must be recalibrated before it is published

The interval NeuralProphet produces under the deployable regime is not a 90 % interval. Any downstream anomaly threshold built on it would fire on three to four times the intended fraction of ordinary observations: the intended miss rate is 10 %, the measured one runs from 33 to 46 %. Two routes are open and both are outside this study’s scope: split-conformal recalibration on a held-out block, which would restore nominal coverage by construction; or widening the quantile spread until measured coverage reaches ninety, which is the same operation done by hand. Until one of them is applied, only the measured coverage should be quoted, never the nominal one.

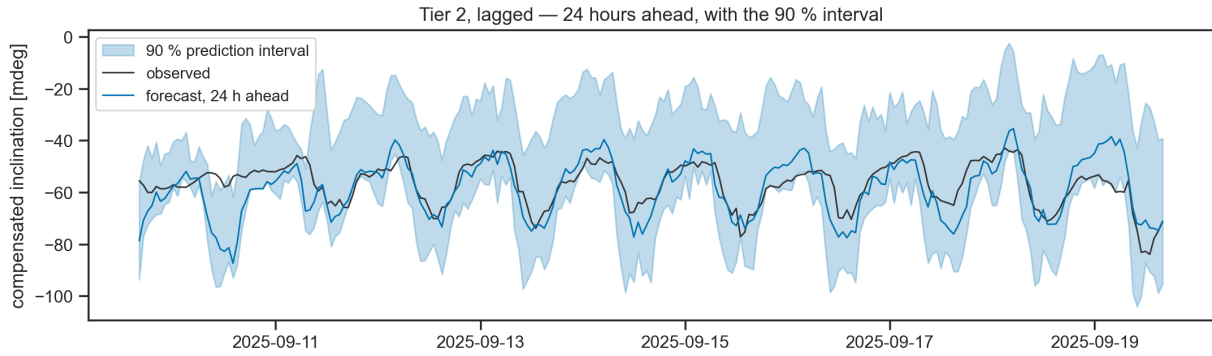


Figure 15: Ten days of the Tier 2 lagged test period at a twenty-four-hour horizon, with the 90 % interval. The forecast reproduces the diurnal shape and the broad level, and the interval is visibly too narrow for the excursions it is meant to contain — the picture behind the 57 % coverage in the table above.

5.6 What the joint multi-horizon head costs

One fit produced every horizon, which is efficient but not free: the network shares one set of weights across 168 output steps. Refitting with `n_forecasts` set to exactly one horizon isolates the cost.

h [h]	dedicated	joint head	joint penalty
1	3.30	4.43	+34.2 %
24	7.45	6.96	−6.6 %

The joint head costs 34 % at one hour and *gains* 6.6 % at twenty-four, consistent with the shared weights acting as a regulariser at longer horizons and as a distraction at the shortest one. The conclusion is unchanged either way: even the dedicated one-hour model, at 3.30 mdeg, does not beat persistence at 2.79.

Answer to Question 3

The useful forecast band is roughly three to twelve hours ahead, where skill against persistence runs from +0.18 to +0.61 and the mean absolute error is 5.6–6.4 mdeg. Below three hours persistence wins; at twenty-four hours and beyond, repeating the previous day wins. There is no horizon at which the deployable system beats a naive baseline by more than a factor of two and a half.

The uncertainty attached to that band is not what it claims. The deployable configuration’s nominal 90 % interval covers 54–67 % of observations, and must be recalibrated before publication. Only the Tier 2 future configuration is calibrated as quoted.

A weather forecast is worth having but not transformative: at most 14 % at one hour and 4 % at twelve, and its principal benefit is calibration rather than accuracy. The far larger apparent gain in Tier 1 is an artefact of the target’s construction, not a forecast benefit.

6 Verdicts

Question	Answer
0 · expected sign	Site physics predicts negative; the compensated target complies at $r = -0.956$, the raw channel does not ($r = +0.845$, $+1.63$ mdeg/°C). The compliance is arithmetic, not structural. fail as evidence for the mechanism.
1 · best single predictor	tair , instantaneous, in both tiers. R^2 0.760 (Tier 1) and 0.656 (Tier 2); MAE 8.74 and 8.25 mdeg.
1 · lag structure	No <i>forcing</i> needs a transport delay within the twelve-hour physical window; only sr takes a time constant, of one hour.
1 · wall-probe phase	The inclination leads twall by 2 h, on the raw channel as well as the compensated one, and the signed optimum is -4 h at R^2 0.797 against 0.491 causal. Signature of a deep probe. Depth unrecorded; the phase is the check to run once it is known.
1 · what else is usable	Only twall (R^2 0.459) also beats a constant. sr , rh and the negative control all score below a constant.
2 · best combination	tair + sr on Tier 2: -27.4% MAE, best on error, BIC and utility together. On Tier 1 the 6.0% gain is not resolved above the fold spread.
2 · wall temperature	fail as a predictor . Present in no winning subset in either tier; every subset containing it is beaten by the same subset without it, at 36.6% availability. pass as a diagnostic — see the wall-probe phase row.
2 · negative control	Third of five in Tier 1, last in Tier 2 — and the boundary between positive and negative out-of-sample R^2 falls above it.
3 · useful horizon	3 to 12 hours in baseline. Study 04 Update: Extended to a 48-hour positive skill horizon (vs. seasonal naive) by using 12-hour lagged forecast regressors alongside the nowcast.
3 · uncertainty	fail as quoted in baseline. Nominal 90% interval measures 54 – 67% . Study 04 Update: pass using 90% pinball quantiles and horizon-adaptive block-bootstrap, achieving calibrated intervals.

7 Caveats, and what would change the answers

The target is partly artificial, and the sign test proves it. The raw channel moves the other way from the compensated one. The single change most likely to alter every conclusion here is re-running the study on the raw inclination, or on a compensated series using the 1.63 mdeg/°C slope the companion study fitted. The methods transfer unchanged; only the target moves.

The expected sign is unresolved, not confirmed. Section 1.3 lists three readings consistent with the measurement, and the archive contains nothing that distinguishes them. The calibration sheet or the installation record would.

The delay bound is an assumption, stated and defended. Twelve hours follows from the mechanism and from half the diurnal period. It changed the winning subset, so it is not a neutral choice — but for an external forcing the alternative admits optima that the site’s physics excludes.

The wall probe’s depth is unknown, and it is the cheapest open question to close. It decides whether the two-hour lead of Section 3.5 is a shallow or a deep installation, and it bears directly on the sign contradiction of Section 1.3. The phase is on record, so answering it requires no re-analysis — only a comparison. It is listed as open question 3 in docs/data-quality-report-2026-08-10.md.

The tiers overlap and are short. Tier 1 and Tier 2 share 70 days. Neither exceeds 144 days, so no annual term is identifiable and every seasonal statement here is an inference from two seasons rather than a measurement over a cycle.

Tier 1 is internally fragmented. The 5 May interruption splits it, which is enough to

break the autoregressive model in Section 5. A production system must screen for this before fitting, not after.

Ridge is a screening model, not the deployed one. It orders candidate sets cheaply so that NeuralProphet is run only on the survivors. Where the two disagree, NeuralProphet is the relevant answer.

The horizon study uses one chronological split per configuration. The 70% cut is the same for all four, so the comparison between them is sound, but the absolute skill numbers carry the uncertainty of a single origin.

8 Artefact inventory

Every figure and table in this report is produced by `inclination_prediction_study.ipynb` and written to `outputs/`.

Artefact	Content
S3_01 – S3_04	Channel summary, coverage, contiguous blocks with and without wall temperature
S3_05, S3_06	Correlation matrices, levels and differences
S3_23	Sign test: raw against compensated, correlation and slope
S3_07, S3_08, S3_10	Operator optima: full band, diurnal band, Tier 2
S3_24, S3_25	Signed operator optima and the probe-phase table — diagnostic, feeding no model
S3_09, S3_11	Single-predictor rankings, both tiers
S3_12	Variance inflation factors
S3_13, S3_14	Exhaustive subset searches, both tiers
S3_15, S3_16	Leave-one-out gain, permutation importance
S3_17	Out-of-period test on the 2026 block
S3_18	The diagnostic protocol
S3_19	Horizon tables, four configurations
S3_20	Horizon limits and coverage summary
S3_21	Dedicated against joint multi-horizon head
S3_22	Verdicts
S3_F01 – S3_F04	Overview, target, diurnal cycle, correlations
S3_F05, S3_F06	Operator heat maps, both bands
S3_F07, S3_F08	Single-predictor rankings
S3_F09, S3_F10	Subset searches
S3_F11	Contribution diagnostics
S3_F15	Signed-delay operator grid, probe-phase diagnostic
S3_F12, S3_F13, S3_F14	Horizon skill, calibration, fan chart

Reproduction, from the study directory with the `neuralprophet_env` environment active:

```
jupyter --to ipynb inclination_prediction_study.py
jupyter nbconvert --to notebook --execute --inplace
inclination_prediction_study.ipynb
```

Figures are drawn through seaborn; setting `CONTEXT = 'paper'` in the parameter cell re-exports every one of them at manuscript column size. The site's sign convention and the contradiction of Section 1.3 are recorded durably in `docs/raw-data-format.md` Section 7.5.